

Exercise 1: SQL Fundamentals

1. SQL: Structured Query language used to communicate with databases.

(0.1) Retrieve all columns from the employees table.

~~id~~ . ~~first name~~ . ~~last name~~ . ~~department~~ . ~~salary~~

id	first name	last name	department	salary	hire date	city
01	Alice	Green	IT	70000	2020-01-10	Johannesburg
02	Brian	Lee	HR	45000	2019-03-22	Cape Town
03	Cathy	Zulu	Finance	65000	2018-07-18	Durban
04	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
05	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

02. All unique departments

SQL: fundamentals logical operators, select column from employee table. select statement allows us to select specific columns in our data.

department.
IT
HR
Finance
Marketing
IT

03 SQL: fundamentals

we want to see ~~last names~~, first names, last names

first name	last name
Alice	Green
Brian	Lee
Cathy	Zulu
David	Mokoena
Eva	Naidoo

04. Top 3 highest-paid employees.

SQL → Structured Query language

• DMS Global system to get data from database.

id	first name	last name	salary
05	Eva	Naidoo	72000
01	Alice	Green	70000
03	Cathy	Zulu	65000

05. SQL fundamentals, general syntax

we want to see ~~an~~ id, ~~names~~, first name, last name, department. from the employees table.

id	first name	last name	department
01	Alice	Green	IT
05	Eva	Naidoo	IT

06. SQL fundamentals, structured Query language

Select statements

↳ allows us to select specific columns in our data

↳ employee table

id	first name	last name	department	salary
01	Alice	Green		70000
03	Cathy	Zulu	Finance	65000
05	Eva	Naidoo		72000

07. SQL: Structured ~~fundamentals~~, ~~stru~~ Query language

We want to see id, first name, last name and department.

id	first name	last name	department
02	Brian	Lee	HR
04	David	Mokoena	Marketing

08. SQL: Structured Query language Used for data processing analysis filtering, aggregation etc.

id	first name	last name	department
02	Brian	Lee	HR
03	Cathy	Zulu	Finance
04	David	Mokoena	Marketing

09. SQL fundamentals logical Operations

and, or, between, not $\leftarrow$ where
in
between

id	first name	last name	department
01	Alice	Green	IT
05	Eva	Naidoo	IT
02	Brian	Lee	HR
03	Cathy	Zulu	Finance

10. SQL Operations. Arithmetic operation

id	first name	last name	department	salary	city
01	Alice	Green	IT	70000	Johannesburg
03	Cathy	Zulu		65000	
05	Eva	Naidoo	IT	72000	Johannesburg

Bright learn Data Analytics / Research Assignment.

1. Database is online transactional processing, information that is capture online while making a transaction. For example the time, ~~per~~ the product information.

Databases are essential because they help to keep record of the transactions and help business in making profitable decisions. Database can be used in ~~se~~ restaurants and imports/exports goods exchange.

2. Data warehouse is where all the information about the transaction is stored, captured. It is where all the records about the transaction are stored. It's different from a regular database because it keeps records of all different areas in one place. Company uses data warehouse to help make profitable decisions ~~to~~ to help the company revenue.

3. Data lake is a centralized area where a vast amount of raw data is stored.

Differences from Data lake and data warehouse.

Data lake
Raw, unprocessed
Schema-on-read
Cheap object storage

Data Warehouse
Processed, refined, structured
Schema-on write (predefined)
Expensive, ~~the~~ high-performance.

Companies ~~at~~ would prefer data lake over ~~rather~~ data warehouse because of large volumes of raw data that needs to be stored.

When you want low-cost, long-term archival before deciding what to use.

4. Data mart is a subset of a data warehouse, focused on a specific business function, (sales, marketing). It contains curated, summarized data.

A data warehouse is the central, enterprise. Data marts are derived from the warehouse or built directly from source systems. They are smaller, faster, and easier to navigate for a particular audience.

Typical users are business analysts and department managers. (Sales, marketing)

5. Structured data; data that is organized neatly in rows, and columns. Easy to search and query with SQL.
Example of structured data; Relational database table with id, names, last names, department, salary.

Semi-Structured data; data that does not fit neatly into a table but contains tags, markers, or a hierarchy to separate elements.

Example of semi-structured data; ("name": "Tijani", "age": "30", "title": "data analyst")

Unstructured data; is data with no particular format or organized. It can not be stored in rows and columns, requires special processing.

Example of unstructured data could be video file, pictures on social media post.

6. Data modelling is the process of creating a visual blueprint or abstract representation of the data an organization uses. Without a model, data becomes chaotic. It forces business stakeholders, developers and analysts to agree on definitions.

It prevents duplicate, orphaned, or invalid data from entering the system.

Two essential data models; The relational model OLTP - Operational systems. It organizes data into strict, two dimensional tables composed of rows and columns.

Dimensional model (OLAP - Analytical systems / data warehouse)

A denormalized model optimized entirely for read-speed and intuitive business reporting.

7. Data mining is the computational process of discovering hidden patterns, anomalies, correlations and predictive trends, within massive datasets by combining statistics, machine.

Businesses use data mining to automate decision making, reduce risk, and maximize customer value.

The data, millions of historical customer transaction receipts.

The mining technique to identify sets of items that statistically co-occur in baskets more often than random chance.

Banking data, streaming transaction data (amount merchant category, location, time, device id, plus).

8. Data management system is where the coding is done (DBMS) Data, Bricks, Microsoft, Snowflake.

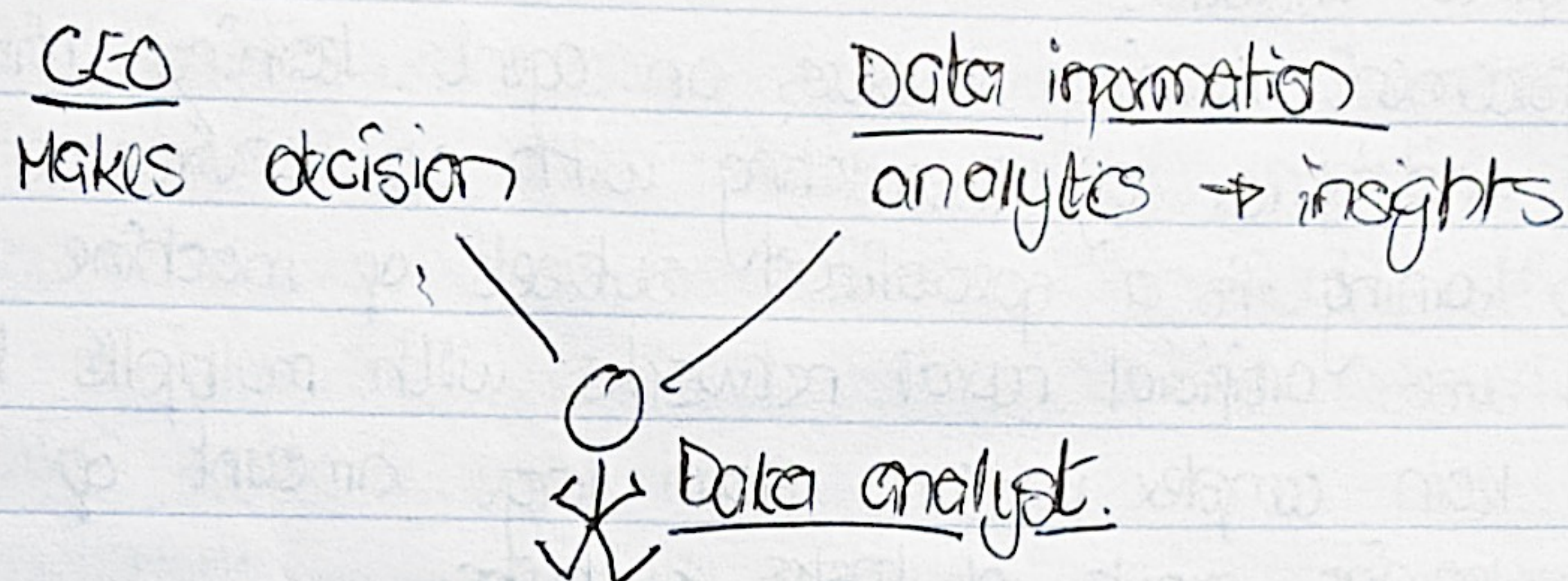
(i) MySQL; Relational Database management system, known for its simplicity and reliability.

(ii) PostgreSQL; Relational Database Management System, an advanced, open-sources that strongly adheres to SQL standard. Allows for custom types, domains and extensions.

(iii) Oracle Database; Converged database, deep integration of AI directly into the database, including support for large language.

(iv) MongoDB; document-oriented NoSQL Database. A leading modern data platform that stores data in flexible.

9. ETL (Extract, Transform, Load); transport data from online transactional processing to (OLAP) online, analytical processing. Data Story telling, You need to be able to tell a story from your data.
You need to drive/extract actionable insights from your data



10. CSV; comma separated values
stores data as plain text with rows and columns
Easy to create, read, and share.
- JSON; JavaScript Object Notation
Stores data in a structured hierarchical format.
 - Avro, row-based storage format
supports schema evolution
 - Parquet are among the most commonly used formats because they provide excellent performance, scalability and compression.
11. Artificial Intelligence (AI) is like a personal assistance that helps you with daily tasks.
Weak AI is for your assistant in definition phrases
Strong AI is for your deeper data information assistance.
12. Machine learning is a ~~type~~ branch of artificial intelligence that enables computer to learn patterns from data and make predictions without being programmed for every task.

Traditional programming write explicit rules and instructions
Data and rules are input into a program.

- Supervised learning uses labelled data, where the correct answer is already known.
- Unsupervised learning, uses unlabelled data. There are no predefined answers.
- Reinforcement learning involves an agent learning through trial and error by interacting with an environment.

13. Deep learning is a specialized subset of machine learning that uses artificial neural networks with multiple layers to learn complex patterns from large amount of data.
Deep learning excels at tasks such as:
Image recognition
Speech recognition
Natural language processing.

14. Natural language processing is a branch of both AI and ~~natural language~~ computational linguistics that enables computers to understand, and work with human language.

15. A large language model is a deep learning model trained on massive amounts of text to understand and generate human language.

16. Generative AI is a branch of AI that creates new content such as text, images, audio, video, and code.
While traditional AI focuses on analysis and prediction,

17. Computer Vision is the branch of AI that enables machines to understand and interpret visual information from images and videos.

Machines see by converting images and videos.

18. Training dataset is the collection of data used to teach a machine model. It contains examples from which the model learns patterns and relationships.
19. An API (Application programming interface) is a set of rules and protocols that allows different software applications to communicate and exchange data with each other. APIs are important; enables communication between different application and systems.
20. An API key is a unique code used to identify and authenticate a user, application, or system when accessing an API. API key verifies who is making the request. Controls access to API services and data. Tracks API usage. How it should be stored and protected, keep it private and never share it publicly.
21. A REST API is a type of web API that follows the principles of REST. It allows different applications to communicate over the internet using standard HTTP methods. REST APIs are widely used because they are simple, scalable and allow systems to exchange data efficiently.
22. JSON (JavaScript Object Notation) is a lightweight text-based format used to store and exchange data between applications.
23. Prompt engineering is the process of designing and refining instructions given to a AI system to produce accurate outputs.